University of California, Berkeley

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EECS Department

GRADUATE STUDENT GENERAL PETITION FORM

TO BE COMPLETED BY STUDENT (PLEASE USE INTERACTIVE FORM)

FROM: Almeda (LAST NAME) Shm (FIRST NAME) Garanganao (MIDDLE) SID# 3036239478

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STATE REQUEST & REASON FOR REQUEST BELOW:

Petition for CS294-164 (Computational Color) taken Fall 2020 for Graphics/HCI Prelim Breadth Requirement
Class Syllabus and Information Available Here: https://tinyurl.com/cs294-164-fa20
Syllabus, Readings, and Course Schedule are attached.

STUDENT SIGNATURE — DATE: 4/26/2021

TO BE COMPLETED BY RESEARCH OR TEMPORARY ADVISOR

COMMENTS: I am supportive of this petition.
The instructor (Ren Ng) should confirm if this course should be on the prelim list. Shm is making very good progress in her research.

RESEARCH/TEMP ADVISOR SIGNATURE — DATE: 4/28/21

TO BE COMPLETED BY DEPARTMENT

DECISION (CIRCLE ONE): **APPROVED** / ~~**NOT APPROVED**~~

CONDITION:

CHAIR OF GRADUATE MATTERS/PRELIM COMMITTEE SIGNATURE — DATE: 4/29/21

STUDENT NOTIFIED? (CIRCLE ONE) YES / NO

DATE

Rev. 6/30/05

CS294 Computational Color - Fall 2020

Instructor: Ren Ng, EECS
Affiliated Faculty: Austin Roorda & Will Tuten, Vision Science
CS294-164 - Fall 2020
Class: Tuesdays 1-4pm online and synchronous. [Class zoom link]
Office Hours: Thursdays 1-2pm online [Office hours zoom link]

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Announcements (updated 8/31/2020):

- The first class is Tuesday 9/1/2020 1-4pm. Zoom link is above.
- I'm excited about the quality of guest lecturers joining us from worldwide this semester -- an upside of the pandemic. Check the schedule link above for guest lecturers.
- If you are interested to enroll in this class, please fill out this application form and attend the first class. Interest from EECS graduate students, Vision Science graduate students, other graduate students, as well as undergrads -- all are welcome.
- Communication will be via Piazza. Please sign up, and see more announcements there: http://piazza.com/berkeley/fall2020/cs294164
- Class will mostly be in zoom, but we will try out a new virtual workspace tool in our first class. Here is a simple test page so you can make sure that your laptop will be able to connect before class: https://ohyay.co/s/test. You will need to sign in with your email address, and recommendation is to use a relatively modern laptop/desktop running Chrome, if possible.
- This is a shortcut for this class webpage: https://tinyurl.com/cs294-164-fa20

Course Description:

- The course will be similar to 2019's offering with improvements. 2019 class materials: http://tinyurl.com/cs294-164

Communication:

- Piazza: http://piazza.com/berkeley/fall2020/cs294164. Please sign up!

Course Meetings:

- Tuesdays, 1-4pm, with mid-way break.

    - This course is online, synchronous, and as a discussion-based course, attendance is essential.
    - See top of this page for zoom link.
    - First half is student-led discussion of the readings
    - Second half is lecture.
    - Course Schedule
- Instructor office hours are Thursdays 1-2pm.
    - See top of this page for zoom link

Grading Policy:

- 25% lead discussion
- 25% weekly reading responses, attendance + class participation
- 50% project.

Expected Course Workload Breakdown

- 4 units = 12 hours per week x 14 weeks
    - 3 hours lecture x 13 weeks
    - 3-6 hours reading x 13 weeks
    - Prepare + lead discussion = 6-9 hours x 1 week
    - Project: 6-9 hours x 12 weeks

Student Deliverables:

- Weekly Responses to Assigned Reading
    - Submit your weekly reading responses by Tuesday 8:59am **before** each class:
    - Examples of good reading responses, though a bit long (from Kyros Kutulakos' class at U. Toronto): example 1, example 2.
- Attendance and Participation
    - This is a discussion-based research seminar. The goal is for a diverse group of students from EECS and Vision Science, and to work to bridge the knowledge and perspectives of the two populations.
- Present Reading and Lead Class Discussion (once per semester, probably in pairs)
    - Prepare a short presentation (10-15 min max) on the assigned reading, distilling the main ideas.
    - Facilitate a group discussion (20 min) about the paper and related ideas.

Projects (draft -- subject to change)

- Semester-long group projects. Projects page.
- See the Course Schedule for timeline of proposal, milestones and final deliverables.
- Final deliverables include a class presentation, 2 minute video, and write-up in journal paper format.

CS294 Computational Color - Readings

Instructor: Ren Ng, EECS
Affiliated Faculty: Austin Roorda & Will Tuten, Vision Science
CS294-164 - Fall 2020

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Check this list regularly for updates.

This reading list is tentative until 1-2 weeks before the relevant meeting. You should be able to download these papers with your Berkeley account. Check with the instructor if you have any problems or questions.

- Meeting 1 - Introduction
  - No required readings; no student-led discussions because first meeting.
  - Optional:
    - Will There Ever Be New Colors That We Can See?
- Meeting 2 Readings for Discussion on September 8 - Tristimulus Theory and Colorimetry
  - Required:
    - Stone, 2005, Representing Colors as Three Numbers.
    - Mancuso et al, 2009, Gene therapy for red–green colour blindness in adult primates.
  - Reference:
    - Koenderink, 2018, Color in The Wild, pp.11-38.
    - Brainard, Stockman, 2010, Colorimetry.
    - CS184 (2020), Lectures 21/22 Video on Color
    - Levoy Intro Applets on Color Theory (Adobe Flash required)
    - Fairchild, Color Appearance Models
    - Hunt, The Reproduction of Color
    - MacAdams, Selected Papers in Colorimetry - Fundamentals
- Meeting 3 - On expanding the dimensionality of color vision
  - Required:
    - Schmidt et al., 2014, Neurobiological hypothesis of color appearance and hue perception.
    - Neitz and Neitz, 2014, Curing Color Blindness—Mice and Nonhuman Primates.

    - Reference:
        - Gómez-Robledo et al. 2018, Do EnChroma glasses improve color vision for colorblind subjects?
        - Neitz & Neitz (2011) - The genetics of normal and defective color vision
        - Graham and Hsia, A unilaterally dichromatic subject (woman with one trichromatic eye and one dichromatic eye)
- Meeting 4 - Oz Vision System Overview
    - Required:
        - Maxwell, 1857, Experiments on Colour, as perceived by the Eye, with Remarks on Colour-Blindness [alternate link]
        - Sabesan et al. The elementary representation of spatial and color vision in the human retina. (web)
    - Reference:
        - Arathorn et al. 2007, Retinally stabilized cone-targeted stimulus delivery
        - Poonja et al. 2005, Dynamic visual stimulus presentation in an adaptive optics scanning laser ophthalmoscope. Per B. Schmidt: "An old, but clearly written paper, describing how a scanning laser system is used to paint a picture on the retina."
        - Rucci et al, 2018, Temporal Coding of Visual Space (web)
    - References related to rendering software for Oz Vision
        - Lian et al. 2019. Ray tracing 3D spectral scenes through human optics models
        - Cook, 1986, Stochastic Sampling in Computer Graphics.
        - A Pixel is Not A Little Square , Alvy Ray Smith, 1995
        - CS184 (2018), Lecture 3 Video on Sampling and Antialiasing
        - The Fourier Transform and It's Applications (Ch. 5) by B. Osgood (this is an outstanding reference)
        - A Video Illustration of Temporal Aliasing. by Jesse Mason
        - Polans et al., Optica, Wide-field optical model of the human eye with asymmetrically tilted and decentered lens that reproduces measured ocular aberrations.
- Meeting 5 - Neural Image Coding
    - Required:
        - Anderson, A.G., Ratnam, K., Roorda, A., and Olshausen, B.A. (2020). High-acuity vision from retinal image motion. J. Vis. 20, 34.
        - [No second paper. Please use piazza to discuss teaming this week if needed, and we will use second paper slot in class for ohyay mingling and team discussion.]
    - Reference:
        - Ruderman et al, 1998, Statistics of cone responses to natural images: implications for visual coding
        - Jacobs - 2014 - The Discovery of Spectral Opponency in Visual Systems and its Impact on Understanding the Neurobiology of Color Vision

        - O'Regan & Noe 2001, A sensorimotor account of vision and visual consciousness
- Meeting 6 - No class discussion (In-Class Pitches instead)
- Meeting 7 - Intro to eye anatomy and visual neurobiology
    - Required:
        - Zhang et al. Cone photoreceptor classification in the living human eye from photostimulation-induced phase dynamics. PNAS (2019). OCT measurements are state-of-the-art for classifying cone spectral type quickly and accurately.
        - Benson, Manning, Brainard. Unsupervised Learning of Cone Spectral Classes from Natural Images. PLoS Comp Biol (2014). Theoretical treatment of how the visual system might uncover the number of cone spectral types (and their cellular arrangement) from responses to natural images; evaluates 2-, 3-, and 4-cone cases.
    - Reference:
        - Mould, How one molecule gives rise to all photoreceptors
        - Solomon & Lennie. The machinery of colour vision. Nat Rev Neurosi (2007)
        - Stockman & Brainard: Color vision mechanisms. (2010)
- Meeting 8 - Color models and camera pipeline
    - Required:
        - Heer, Stone, 2012, Color Naming Models for Color Selection, Image Editing and Palette Design
        - Gharbi, Chaurasia, Paris, Durand, 2016, Deep Joint Demosaicking and Denoising.
    - Reference:
        - Koenderink, 2018, Color in The Wild, remaining pages not read earlier.
        - G. Agranov, V. Berezin, R.H. Tsai, 2003, Crosstalk and Microlens Study in a Color CMOS Image Sensor.
        - McCamy, Marcus, Davidson. A Color-Rendition Chart. 1976. Paper introducing the MacBeth Color Chart commonly used in color reproduction applications.
        - Wright. A re-determination of the trichromatic coefficients of the spectral colours,1929. Experimental color matching on 10 people, in conjunction with work by Guild on another 7, used as basis for CIE 1931 RGB color space, defined as 2 degree “standard observer” definition.
        - Guild, The colorimetric properties of the spectrum, 1931. Experimental color matching data on 7 people, in conjunction with work by Wright on 10, used as basis for CIE 1931 RGB color space, defined as 2 degree “standard observer” definition.
        - Wolf, S. 2003. NTIA Technical Memo TM-04-406 Color correction matrix for digital still and video imaging systems
- Meeting 9 - Color in Practice (from the perspective of a photographer)

- Required:
  - Thein, blog articles on photography (read these four articles)
    - The Four Things https://blog.mingthein.com/2016/07/28/the-four-things-redux/
    - Structure https://blog.mingthein.com/2019/09/02/back-to-basics-structure/
    - Understanding Color from a Workflow Perspective https://blog.mingthein.com/2018/11/24/understanding-color-from-a-workflow-perspective-1/ https://blog.mingthein.com/2018/11/27/understanding-color-from-a-workflow-perspective-part-2/
  - Barron, 2015, Convolutional Color Constancy
- Reference:
  - The Four Things https://blog.mingthein.com/2016/07/28/the-four-things-redux/
  - Turning an idea into an image: https://blog.mingthein.com/2019/09/14/back-to-basics-turning-an-idea-into-an-image/
  - Structure: https://blog.mingthein.com/2019/09/02/back-to-basics-structure/
  - Layering: https://blog.mingthein.com/2019/09/08/back-to-basics-layering/
  - Subject isolation: https://blog.mingthein.com/2019/08/27/back-to-basics-cut-points-and-edges/
  - Rules of vision part I: https://blog.mingthein.com/2019/08/09/back-to-basics-rules-of-vision-or-things-we-cant-help-seeing-part-i/
  - Rules of vision part II: https://blog.mingthein.com/2019/08/15/rules-of-vision-part-ii/
  - Viewing and presentation methods: https://blog.mingthein.com/2018/03/14/on-viewing-and-presentation/
  - Practical differences between eye and camera: https://blog.mingthein.com/2018/06/03/repost-practical-differences-between-cameras-and- human-vision/
  - Understanding HDR, the zone system and dynamic range: https://blog.mingthein.com/2015/08/31/repost-hdr-the-zone-system-and-dynamic- range/
  - Curation https://blog.mingthein.com/tag/curation/
  - The medium format look https://blog.mingthein.com/2015/02/24/that-medium-format-look-what-is-it/ https://blog.mingthein.com/2019/07/16/what-defines-small-medium-large-formats-anyway/

        - Limitations of language https://blog.mingthein.com/2020/07/10/limitations-of-language/
- Meeting 10 - No class discussion (In-Class Mid-Point Demos instead).
- Meeting 11 - Field Guide for Hacking the Human Visual System
    - Required:
        - Roorda, A., Adaptive optics for studying visual function: A comprehensive review, Journal of Vision 11(5) article 6 (2011).
        - Chapter 17 in: J. Bille (ed), High Resolution Imaging in Microscopy and Ophthalmology
    - Reference:
        -
- Meeting 12 - Tetrachromacy
    - Required:
        - Jordan et al. 2010. The dimensionality of color vision in carriers of anomalous trichromacy.
        - Jameson, Highnote, Wasserman. Richer color experience in observers with multiple photopigment opsin genes
    - Reference:
        - Webpage of selected papers by Kimberly Jameson.
- Meeting 13 - Color Appearance Models
    - Required (one presenter only):
        - Asano, Fairchild, Blondé, Individual Colorimetric Observer Model, 2016.
    - Reference:
        - Mark Fairchild, Color Appearance Models, 3rd edition, Wiley 2013. Spreadsheet with example calculations (Excel).
        - CIECAM02 example forward and reverse calculations (Excel). Contributed by Eric Walowit and Grit O'Brien from Mark Fairchild's website.

Everything below this line is subject to change!

- Other reference reading:
    - Auto white balance in camera pipelines:
        - Brainard, Wandell, 1986. Analysis of the retinex theory of color vision [alternative]
        - Hui et al. 2018. Illuminant Spectra-based Source Separation Using Flash Photography
        - Barron, 2015, Convolutional Color Constancy

- Color in computer vision:
  - Khan et al., 2018, Modern Trends in Hyperspectral Image Analysis: A Review
  - Flachot and Gegenfurtner, 2018, Processing of chromatic information in a deep convolutional neural network
  - Rafegas and Vanrell, 2018, Color encoding in biologically-inspired convolutional neural networks
  - Emma Alexander's lecture slides on Why See in Color?
- Color in Data Science:
  - Szafir, Stone, Gleicher, 2014, Adapting Color Difference for Design
  - Heer, Stone, 2012, Color Naming Models for Color Selection, Image Editing and Palette Design
  - Stone, Szafir, Setlur, 2014, An Engineering Model for Color Difference as a Function of Size
  - Szafir, 2017. Modeling Color Difference for Visualization Design
  - Stone, 2016. How we designed the new color palettes in Tableau 10

HOME	Student Disc. Topics | 1:10 - 2:20	Student Disc. Leaders (See Readings)	Lecture Topic | 2:30 - 4:00	Lecturer	Project Deadlines
1 Sep-1	None - course introduction		Tristimulus theory and colorimetry	Ng	
2 Sep-8	Tristimulus theory and colorimetry	Vivrekar; Almeda	On expanding the dimensionality of color vision	Jay Neitz	
3 Sep-15	On expanding the dimensionality of color vision	Ng; Petryk	Oz Vision overview	Ng	
4 Sep-22	Oz Vision overview*	Herbeck; Qiao	Neural image coding*	Bruno Olshausen	
5 Sep-29	Neural image coding	Cai; [ohyay team discussion]	Intro to eye anatomy and visual neurobiology	William Tuten	Choose team; project proposal
6 Oct-6	None - In-Class Pitches		Color in practice (data science perspective)	Maureen Stone	In-class pitches
7 Oct-13	Guest lecture: Color processing in the camera pipeline	Guest lecturer: Michael Brown	3:00pm: Color in practice (graphics perspective)	Masha Shugrina	
8 Oct-20	Color in data science and in the camera pipeline	Maudgalya; Liu, W.	Color in practice (photographer perspective)	Ming Thein	
9 Oct-27	Color in practice (photographer perspective)	Hu; Xu	Field guide for hacking the human visual system	Austin Roorda	
10 Nov-3	None - In-class demos / midpoint presentations		Optional: office hours in ohyay/zoom		
11 Nov-10	Single-photoreceptor stimulation hardware*	Wongpanich; Upadhyay	Color dimensionality and tetrachromacy*	Jenny Bosten	Check-in report #1
12 Nov-17	Color dimensionality and tetrachromacy*	Raghavan; Chen	Color appearance models*	Mark Fairchild	
13 Nov-24	Color Appearance Models	Tang;	Next-gen single-photoreceptor imaging and stimula	Ramkumar Sabesan	
14 Dec-1	**Final Presentations**				Final presentations
Dec-8					Report + video due
	No Final Exam				

(*) On these dates the guest lectures will be 1:10-2:40pm, and the student discussion will be 2:50 - 4:00pm. This is to accommodate guest lecturer schedule constraints.